PROBABILITY THEORY · CHAPTER 2

Discrete Random Variables

A friendly, worked-example study guide — what each idea means, how the pieces connect, and exactly how to think when a question lands in front of you.

WHAT'S INSIDE

Covers slide sets P2.0 – P2.4 · Built as a companion to your lecture notes

0 The big picture

READ THIS FIRST — IT TIES THE WHOLE CHAPTER TOGETHER

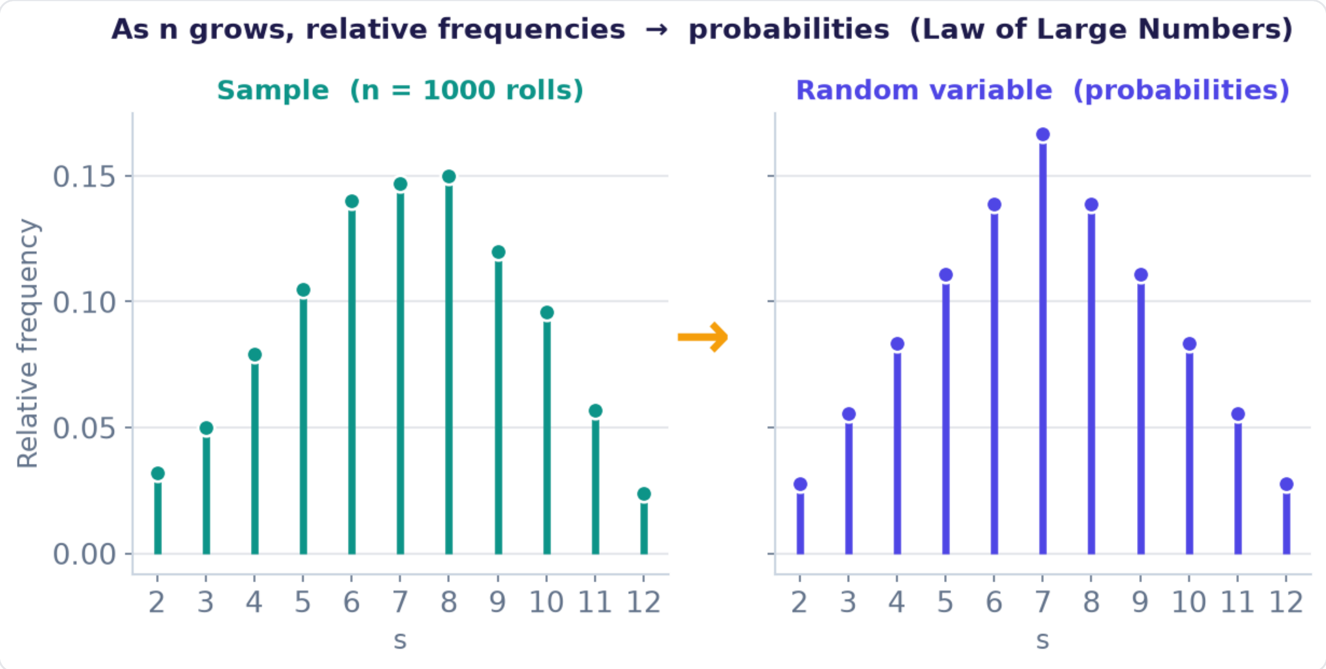
Everything in this chapter is one idea seen from two sides: **descriptive statistics describes the data you actually collected; probability theory describes the whole population the data came from.** The bridge between them is the law of large numbers.

In descriptive statistics you had a **sample**: you counted outcomes, made a frequency table, drew a needle graph, and computed a mean and standard deviation. In probability theory you do the exact same things — but for the idealised population, where "relative frequency" becomes "probability". The vocabulary changes; the intuition does not.

THE ONE TABLE THAT MAPS THE WHOLE CHAPTER

Descriptive statistics (sample)	Probability theory (population / random variable)	Where in this guide
A DATA VALUE	An outcome m_i of a random variable X	Part 1
RELATIVE FREQUENCY OF A VALUE	Probability $P(X=m_i)$	Part 2
FREQUENCY TABLE + NEEDLE GRAPH	Probability distribution + needle graph	Part 2
CUMULATIVE RELATIVE FREQUENCY	Cumulative distribution $F(x) = P(X \le x)$	Part 3
SAMPLE MEDIAN, QUARTILES	Population median, quantiles / percentiles	Part 3
SAMPLE MEAN $\bar{x}$	Population mean $\mu = E(X)$	Part 4
SAMPLE VARIANCE / ST.DEV. s	Population variance σ^2, st.dev. σ	Part 4

Why are the two sides linked at all? Because of the **law of large numbers**: if you repeat a random experiment more and more times, the relative frequency of each outcome settles down towards a fixed number — and that number is what we call its probability.



Roll two dice 1000 times (left): the relative frequencies already look almost identical to the true probabilities (right). Let $n\to\infty$ and they coincide. This is why a probability distribution is just the "infinite-sample" version of a frequency table.

HOW TO THINK

Whenever a new symbol appears ($E(X)$, σ^2, $F(x)$, a quantile…), ask: "what was the sample version of this in descriptive statistics?" The population formula is almost always the sample formula with **relative frequency swapped for probability**. If you know one side, you know the other.

1 Random variables

SLIDE SET P2.0 — THE VOCABULARY EVERYTHING ELSE IS BUILT ON

DEFINITION

A **random variable** maps the outcome of a random experiment to a **number**. We name random variables with capital letters (X, Y, T, …); a specific value it can take is written with a small letter (x).

The word "variable" is doing a lot of work: a random variable is not a fixed number, it is a rule that turns "whatever happens" into a number. Think of it as a machine — you feed in the result of an experiment, and out comes a value.

WORKED EXAMPLES OF RANDOM VARIABLES

- T = minutes you wait for a metro that comes every 10 min → any value in [0, 10].
- I = "important to be rich" on a 1–6 agreement scale (European Social Survey).
- A = age in years of a randomly chosen person living in Romania.
- X = social trust on a 0–10 scale for a random European citizen.
- S = sum of the results when rolling two dice.

When the experiment isn't numeric

Many experiments have **non-numeric** outcomes — yet the interesting question is usually still about a number. The trick is to define a new variable that counts or summarises the non-numeric result.

- Opinion on the death penalty: $\Omega = \{\text{pro, against}\}$. Interesting variable: D = number of people against in a sample of 1000. That is a number.
- Rolling two dice: the raw outcome is a pair like (3,5), but S = their sum is the number we study.

WATCH OUT — CODING IS NOT MEASUREMENT

Sometimes categories are turned into numbers by **coding**: gender G = 1 for female, 0 for male; employment status E = 1 employee, 2 worker, 3 self-employed. These digits are labels, not quantities. **Never do arithmetic with coded categories** — an "average employment status of 1.8" is meaningless. (Coding as 0/1 for a genuine two-outcome count, like the Bernoulli variable in Part 5, is legitimate — there the 1 really means "one success".)

Discrete vs. continuous — and why it matters

This single distinction decides which techniques you use for the rest of the chapter.

DISCRETE

Outcomes you can **list** (often whole numbers). Examples: I (1–6), S = dice sum, D = a count. We use **probability distributions and sums (Σ)**. This whole guide is about the discrete case.

CONTINUOUS

Outcomes fill an **interval**; you cannot list them. Examples: T = waiting time, X = social trust on [0,10]. These use density curves and integrals (a later chapter).

HOW TO THINK — STARTING ANY PROBLEM

1. **Name the experiment** in one sentence ("pick a random household in region A").
2. **Define the random variable** precisely, with a capital letter and its unit ("X = number of cars in that household").
3. **Decide discrete or continuous.** Can you list the outcomes? → discrete → you're in this guide.

Doing this out loud fixes 90% of confusion before any calculation starts.

2 Probability distribution & probabilities

SLIDE SET P2.1 — LISTING THE OUTCOMES AND THEIR CHANCES

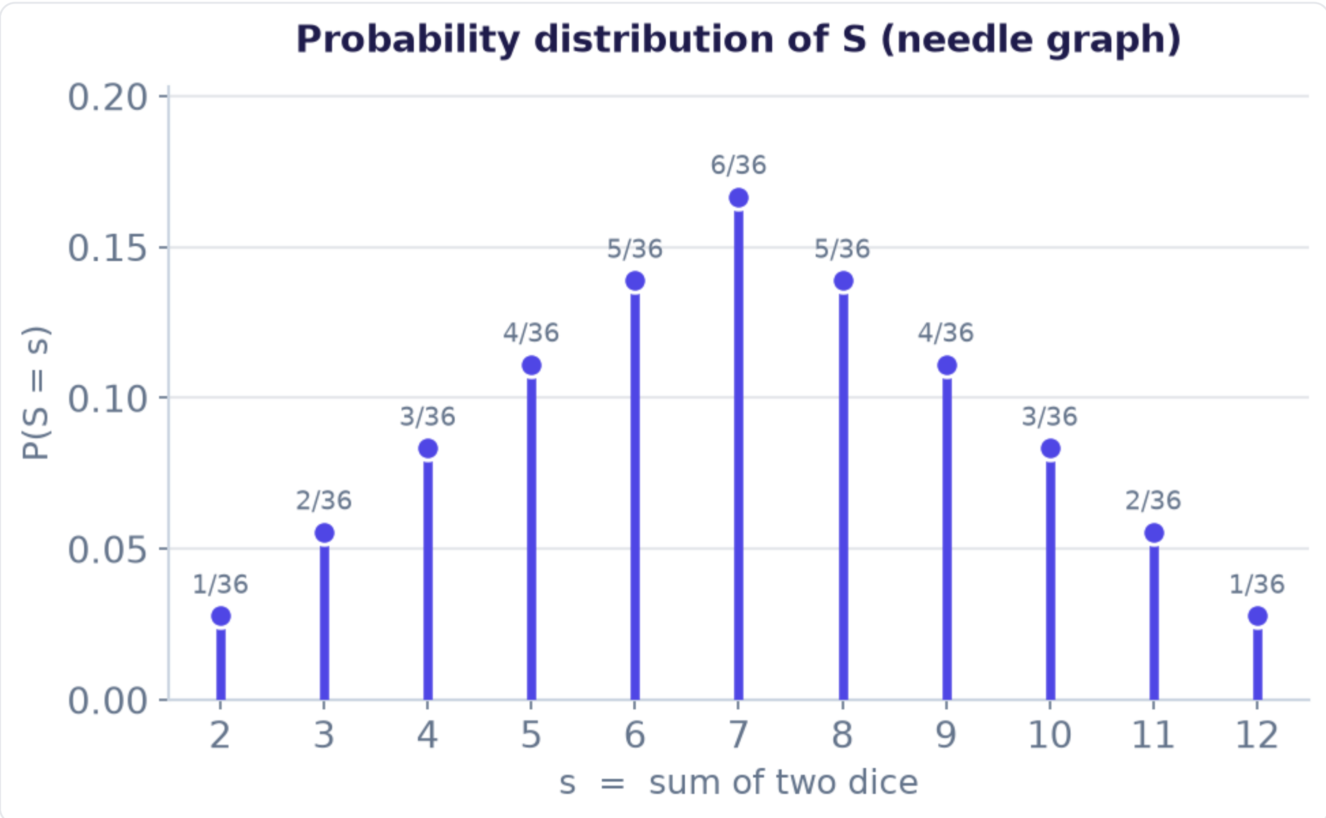
A **probability distribution** is the population twin of a frequency table: it lists every possible outcome next to the probability of that outcome.

THE THREE RULES A PROBABILITY DISTRIBUTION MUST OBEY

1. The listed outcomes are **disjoint** (no overlap — each result counts once).
2. Every probability is **between 0 and 1**: $0 \le P(X=m_i) \le 1$.
3. The probabilities **add up to exactly 1**: $\Sigma\, P(X=m_i) = 1$.

Rule 3 is your best friend in exams: a "complete the table" question is almost always solved by making the column total 1.

Graphically we draw a **needle graph** (a "stick" of height = probability above each outcome) — identical in spirit to the sample needle graph, but the heights are probabilities.



Probability distribution of S = sum of two dice. Each needle's height is the probability of that sum. Note the symmetric triangular shape, peaking at 7 (the most likely sum).

Where do the probabilities come from?

- **Rules of probability** — e.g. Laplace: (favourable outcomes)/(total equally-likely outcomes). For two dice there are 36 equally likely pairs, so $P(S=5) = 4/36$.

- **Standard models** — some situations always follow the same pattern (Bernoulli, binomial — Part 5). Recognise the pattern and the probabilities are handed to you by a formula.
- **Historical data** — estimate probabilities from observed relative frequencies, justified by the law of large numbers.

Probabilities of events (more than one outcome)

An **event** can bundle several outcomes ("one or two cars", "$Y \le 3$"). Because the outcomes are disjoint, the probability of the event is just the **sum** of the individual outcome probabilities it contains.

WORKED EXAMPLE — CARS IN A HOUSEHOLD (OLD EXAM QUESTION)

X = number of cars in a household in region A:

X	0	1	2	3	4
$P(X=X)$	0.15	0.40	0.35	0.09	?

(a) Complete the table. The probabilities must total 1, so
$P(X=4) = 1 - (0.15 + 0.40 + 0.35 + 0.09) = 1 - 0.99 =$ **0.01**.

(b) Probability of one or two cars. These are disjoint outcomes, so add:
$P(X=1 \text{ or } X=2) = 0.40 + 0.35 =$ **0.75**.

WORKED EXAMPLE — RUSSIAN ROULETTE (BUILDING A DISTRIBUTION FROM SCRATCH)

A revolver has 6 chambers, one loaded. Spin, fire, repeat with the same single round until it goes off. Let Y = number of trials until the fatal shot.

To reach trial k, the first $k-1$ pulls must be "click" (prob 5/6 each) and the k-th must fire (prob 1/6). Independent, so multiply:

$$P(Y = k) = \frac{5}{6}^{k-1} \cdot \frac{1}{6} \quad \text{for } k = 1, 2, 3, \ldots$$

This is a geometric-type distribution — the outcomes never stop (1, 2, 3, …), but the probabilities shrink fast and still sum to 1. We'll compute "at most / between" probabilities for it in Part 3.

HOW TO THINK — "FIND THE PROBABILITY THAT…"

1. **Translate the words into outcomes.** "One or two" = {1, 2}; "at least 3" = {3, 4, …}.
2. **Are the outcomes disjoint?** For a single random variable they always are — so you may simply **add** their probabilities.
3. **Missing probability?** Reach for "everything sums to 1".
4. **Long tail / "at least"?** Use the complement: $\mathbb{P}(\text{event}) = 1 - \mathbb{P}(\text{opposite})$. Far less arithmetic.

3 Cumulative distribution & percentiles

SLIDE SET P2.2 — PROBABILITIES OF "UP TO AND INCLUDING"

DEFINITION — CUMULATIVE DISTRIBUTION FUNCTION (CDF)

$$F(x) = P(X \leq x) = \text{sum of all probabilities for outcomes} \leq x$$

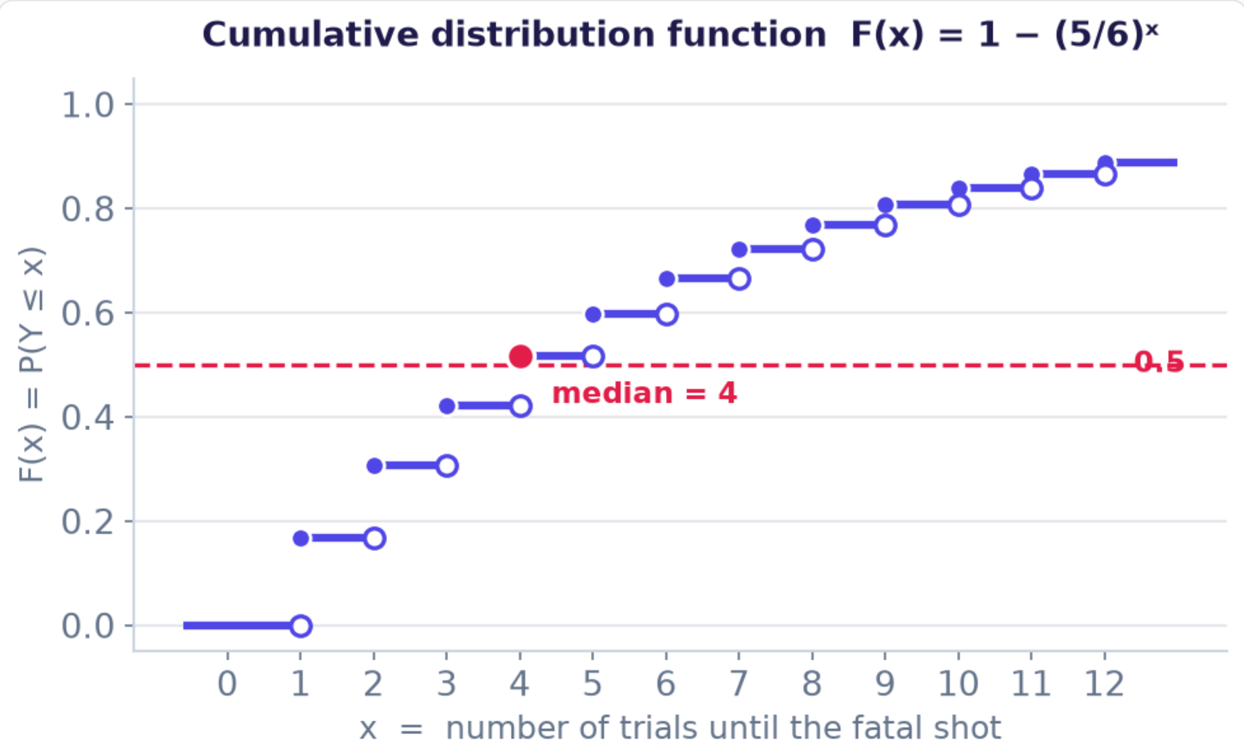
It is the population version of the **cumulative relative frequency**. F starts at 0, climbs by a jump at each possible outcome, and ends at 1.

For a discrete variable the CDF is a **step function**: flat between outcomes, then it jumps up by $P(X{=}m_i)$ at each outcome. The step convention matters: at an outcome the value includes that outcome (the "≤"), so the dot is filled on the left of each new step.

WORKED EXAMPLE — CDF OF THE RUSSIAN-ROULETTE VARIABLE

$F(x) = P(Y \leq x)$. Instead of adding many terms, use the complement — surviving all of the first x pulls has probability $(5/6)^x$:

$$F(x) = P(Y \leq x) = 1 - P(Y > x) = 1 - \frac{5}{6}^{x}$$



The CDF climbs in steps and levels off toward 1. Reading it: the height at any x is the probability the fatal shot has already happened by trial x.

Reading interval probabilities off the CDF

Once you have F, "between" questions become subtractions. The only subtlety is whether each end is included (≤) or excluded (<). For a variable on 1, 2, 3, …:

$$P(a < Y \le b) = F(b) - F(a)$$

WORKED — USING F FOR THE ROULETTE VARIABLE

Question	Rewrite with F	Value
$P(2 < Y \le 4)$	$F(4) - F(2) = (1-(5/6)^4) - (1-(5/6)^2)$	≈ 0.517 − 0.306 = **0.211**
$P(Y \le 2)$	$1 - (5/6)^2$	≈ **0.306**
$P(Y \le 3)$	$1 - (5/6)^3$	≈ **0.421**
$P(Y \le 20)$	$1 - (5/6)^{20}$	≈ **0.974**

The hint made explicit: for large upper limits, don't add twenty terms — use $P(Y \le 20) = 1 - P(Y > 20) = 1 - (5/6)^{20}$. Recognising the complement is the whole skill here.

Percentiles, quantiles & the population median

Quantiles are the population counterparts of the sample median and quartiles — again, take the sample idea and let n grow.

DEFINITION — POPULATION MEDIAN (DISCRETE)

The median is the **smallest outcome x whose cumulative probability reaches at least 0.5**: the smallest x with $F(x) = P(X \le x) \ge 0.5$. More generally the quantile $Q(p)$ is the smallest x with $F(x) \ge p$.

WORKED — MEDIAN OF THE ROULETTE VARIABLE, Q(0.5)

Find the smallest x with $F(x) \ge 0.5$. From the table above, $F(3) \approx 0.421$ (not yet 0.5) but $F(4) \approx 0.517$ (≥ 0.5). So the first time the curve reaches the 0.5 line is at $x = 4$ → **median = 4 trials**. On the graph in the previous section this is exactly where the red dashed 0.5 line meets the steps.

HOW TO THINK — CDF & QUANTILE QUESTIONS

- **"≤" in the question → think F directly.** F already is "≤".
- **"Between a and b" → subtract two F values.** Then double-check each endpoint's < vs ≤ to see if it's included.
- **Big "≤ k" or any "≥"/">" → use the complement** $1 - \mathrm{F}(\dots)$.
- **Median / quantile → walk the cumulative column** and stop at the first outcome that reaches the target (0.5 for the median). "First to reach", not "closest to".

4 Mean & standard deviation

SLIDE SET P2.3 — THE CENTRE AND THE SPREAD OF A RANDOM VARIABLE

Population mean = expected value

DEFINITION — EXPECTED VALUE E(X), THE POPULATION MEAN

$$\mu = E(X) = \Sigma\, m_i \cdot P(X=m_i)$$

The expected value is a **weighted average of the outcomes**, where each outcome is weighted by its probability. It is the population twin of the sample mean $\bar{x}$ — replace "relative frequency of m_i" with "probability of m_i".

"Expected" is a slightly misleading word: $E(X)$ need not be a value X can actually take (the expected dice sum is 7, which is possible; the expected number of cars below is 1.41, which is not a possible count). It is the long-run average, not a prediction of any single trial.

WORKED — EXPECTED SUM OF TWO DICE

Multiply each outcome by its probability and add. The row $s{\cdot}P(S{=}s)$ totals 252/36:

$$E(S) = \frac{2+6+12+20+30+42+40+36+30+22+12}{36} = \frac{252}{36} = \mathbf{7}$$

Sanity check: the distribution is symmetric around 7, and for a symmetric distribution the mean sits at the centre. ✓

WORKED — MEAN NUMBER OF CARS

Using the completed table (0.15, 0.40, 0.35, 0.09, 0.01):

$$E(X) = 0(0.15) + 1(0.40) + 2(0.35) + 3(0.09) + 4(0.01) = \mathbf{1.41\ cars}$$

Population variance & standard deviation

DEFINITION — VARIANCE VAR(X) AND STANDARD DEVIATION

$$\mathrm{Var}(X) = \sigma^2 = E[(X-\mu)^2] = \Sigma\,(m_i-\mu)^2 \cdot P(X=m_i) \qquad \sigma = \sqrt{\mathrm{Var}(X)}$$

Variance = the **average squared distance from the mean**, weighted by probability. The standard deviation σ is its square root, back in the original units — so it's the number you actually interpret ("typical distance from the mean").

THE CONVENIENCE FORMULA — USE THIS IN PRACTICE

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2$$

Same answer, far less work: instead of computing every $(m_i - \mu)^2$, you only need $E(X^2) = \Sigma\, m_i^2 \cdot P$ and the mean you already found. **Careful:** it is "mean of the squares minus square of the mean" — $E(X^2) \neq [E(X)]^2$. Mixing these up is the single most common variance mistake.

WORKED — VARIANCE OF THE DICE SUM, TWO WAYS

Definition route: weight each $(s-7)^2$ by its probability:

$$\mathrm{Var}(S) = \frac{25+32+27+16+5+0+5+16+27+32+25}{36} = \frac{210}{36} \approx \mathbf{5.83}$$

Convenience route: first $E(S^2) = \Sigma\, s^2 \cdot P = 1974/36 \approx 54.83$, then subtract 7^2:

$$\mathrm{Var}(S) = E(S^2) - [E(S)]^2 = 54.83 - 7^2 = 54.83 - 49 \approx \mathbf{5.83}\ ✓$$

Both give 5.83, so $\sigma = \sqrt{5.83} \approx 2.42$.

WORKED — STANDARD DEVIATION OF THE NUMBER OF CARS

We know $\mu = 1.41$. Compute $E(X^2)$:

$$E(X^2) = 0^2(0.15)+1^2(0.40)+2^2(0.35)+3^2(0.09)+4^2(0.01) = 0+0.40+1.40+0.81+0.16 = 2.77$$

$$\mathrm{Var}(X) = 2.77 - 1.41^2 = 2.77 - 1.9881 \approx 0.782 \;\rightarrow\; \sigma = \sqrt{0.782} \approx \mathbf{0.88\ cars}$$

HOW TO THINK — MEAN/VARIANCE QUESTIONS

1. **Build a working table** with a row for $m_i \cdot P$ and a row for $m_i^2 \cdot P$. Summing those rows gives $E(X)$ and $E(X^2)$ in one sweep.
2. **Always use the convenience formula** for variance unless the mean is a clean integer.
3. **Report σ, not σ^2,** when the question asks for "spread" or "standard deviation" — and keep the units.
4. **Sanity-check:** the mean must lie inside the range of outcomes; σ should be comparable to a typical gap from the centre. If σ comes out negative or huge, recheck the convenience formula's order of operations.

5 The binomial distribution

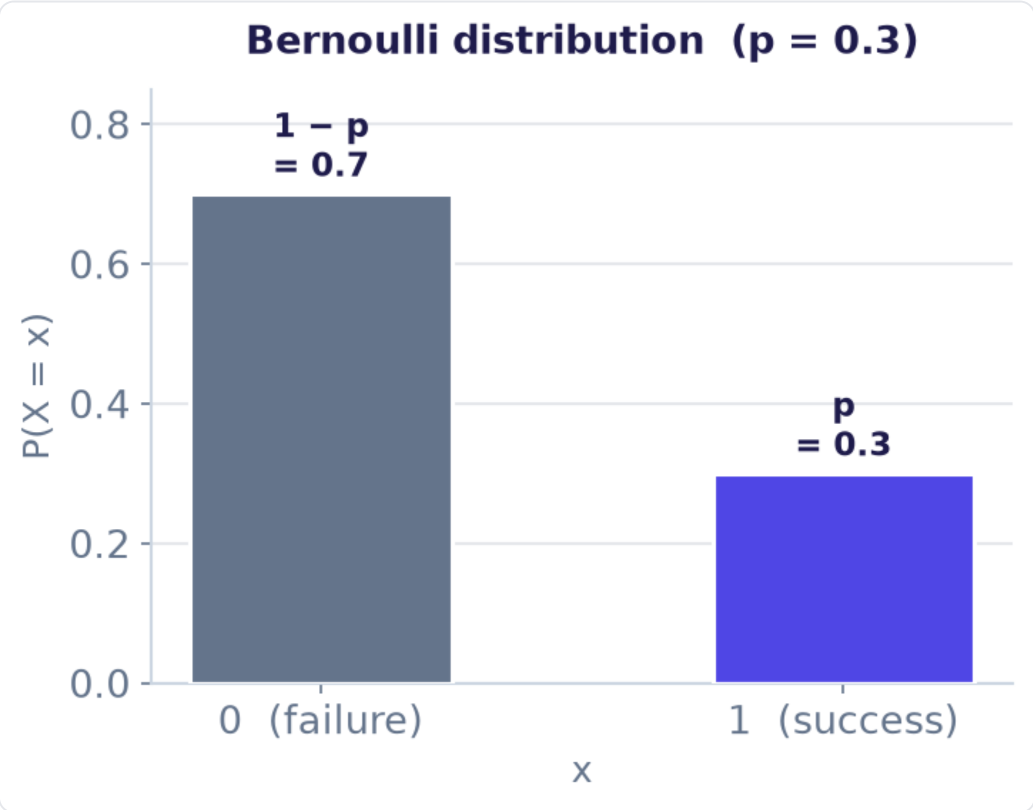
SLIDE SET P2.4 — THE MOST IMPORTANT NAMED MODEL IN THE COURSE

The binomial is the "count the successes" distribution. Whenever you repeat the same two-outcome trial a fixed number of times and count how often one outcome happens, it's binomial — and a single formula hands you every probability.

Step 1 — the building block: the Bernoulli trial

BERNOULLI EXPERIMENT & BERNOULLI VARIABLE

A **Bernoulli experiment** has exactly two outcomes, labelled success and failure. A **Bernoulli variable** X takes the value **1** for success and **0** for failure, with $P(\text{success}) = p$.



A Bernoulli variable: all the probability sits on 0 and 1. Here $p = 0.3$.

Its mean and variance drop straight out of the definitions and become the seeds for the binomial:

$$E(X) = 0(1-p) + 1(p) = \mathbf{p} \quad \cdot \quad Var(X) = E(X^2) - [E(X)]^2 = p - p^2 = \mathbf{p(1-p)}$$

Step 2 — the binomial distribution

DEFINITION

X = the number of successes when a Bernoulli trial with success probability p is repeated n **independent** times. We write:

$$X \sim \mathrm{Bin}(n, p)$$

THE FOUR CONDITIONS — CHECK ALL FOUR (MNEMONIC: B-I-N-S)

- **B**inary — each trial is success/failure.
- **I**ndependent — trials don't affect each other.
- **N** fixed — the number of trials is decided in advance.
- **S**ame p — success probability is identical every trial.

WORKED — "BINOMIALLY DISTRIBUTED OR NOT?"

Scenario	Binomial?	Why
Number of '6's when rolling a die 50 times	Yes	Bin(50, 1/6): binary (6 / not-6), independent, n fixed, same p.
Number of club cards when drawing 5 from a deck	No	Drawing without replacement → p changes each draw, trials not independent.
Number of blue cars passing in the next hour	No	No fixed number of trials n — cars arrive an unknown number of times.
Number against death penalty out of 100 people	Yes	Bin(100, p): 100 fixed trials, each pro/against, ~independent, same p.

The two "No" cases are the classic traps: sampling without replacement breaks independence, and "in the next hour" has no fixed n.

Step 3 — the probability formula, and where it comes from

Take Bin(100, 0.25) as an example. What's the probability of a specific sequence with exactly two successes, say 1 1 0 0 … 0? By independence, multiply: $p^2 (1-p)^{98}$. Any other specific two-success sequence (0 1 0 … 1) has the same probability. So we just need to count how many such sequences exist — that's "choose which 2 of the 100 positions are the successes", the binomial coefficient. Multiply the count by the shared probability:

BINOMIAL PROBABILITY (PROBABILITY MASS FUNCTION)

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n$$

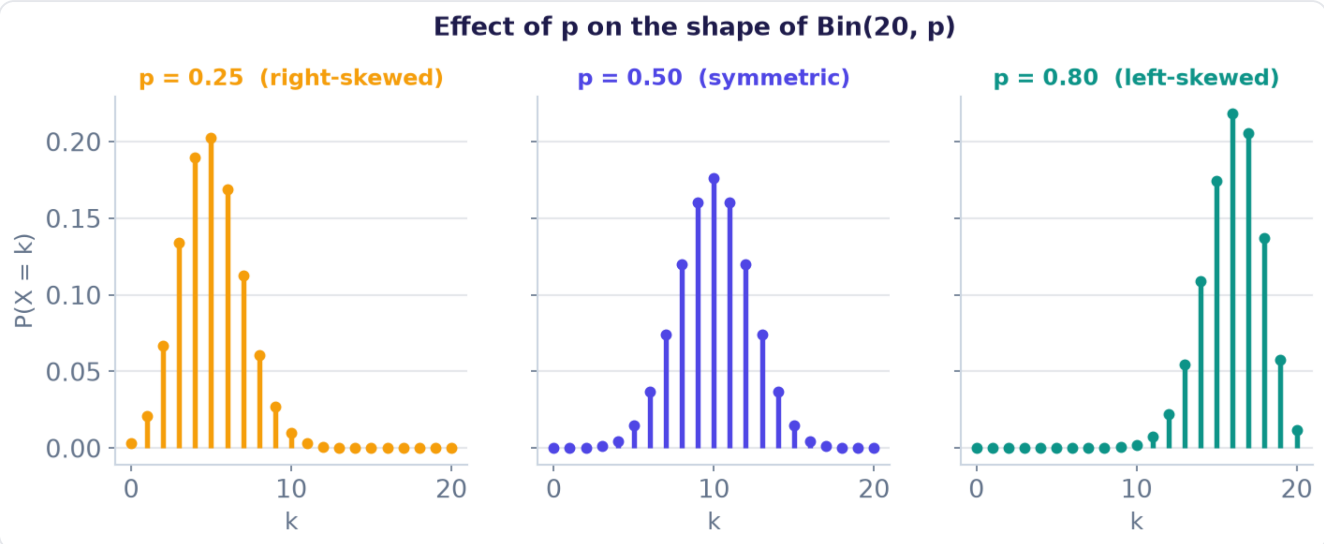
$\binom{n}{k}$ = number of ways to place the k successes · p^k = the k successes · $(1-p)^{n-k}$ = the $n-k$ failures

Step 4 — mean and variance (the shortcut)

Plugging the pmf into the Σ-definitions is painful. The clean way: a binomial variable is just a sum of n independent Bernoulli variables, $X = X_1 + \ldots + X_n$. Means always add, and variances of independent variables add — so:

$$E(X) = np \quad \cdot \quad Var(X) = np(1-p) \quad \cdot \quad \sigma = \sqrt{np(1-p)}$$

Step 5 — the shape of the distribution



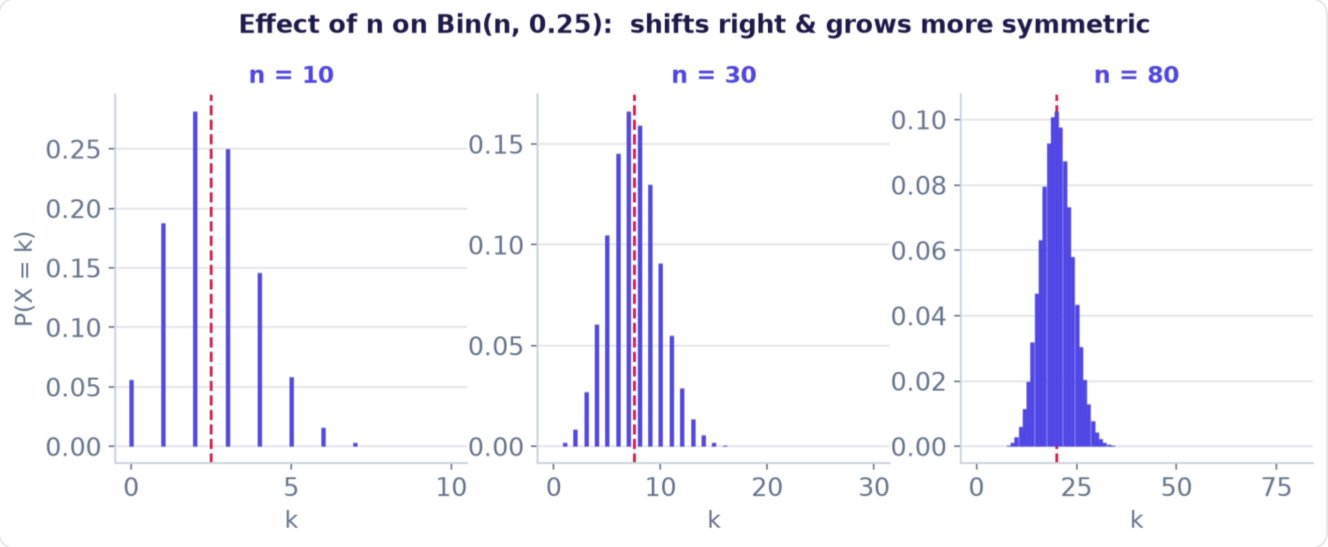
Effect of p (with n = 20 fixed).

Effect of p:

- $p = 0.5$ → **symmetric**.
- $p < 0.5$ → **right-skewed** (tail to the right).
- $p > 0.5$ → **left-skewed**.
- The closer p is to 0 or 1, the more skewed.

Effect of n:

- As n grows the whole distribution **shifts right** (mean np grows) and becomes **more symmetric** — even when p is away from 0.5.
- If p is very close to 0 or 1 a very large n is needed before it looks symmetric.
- For large n the binomial is well approximated by the **normal** distribution (the Central Limit Theorem, later).



Effect of n (with $p = 0.25$ fixed); the dashed line marks the mean np.

Step 6 — computing binomial probabilities (Excel & calculator)

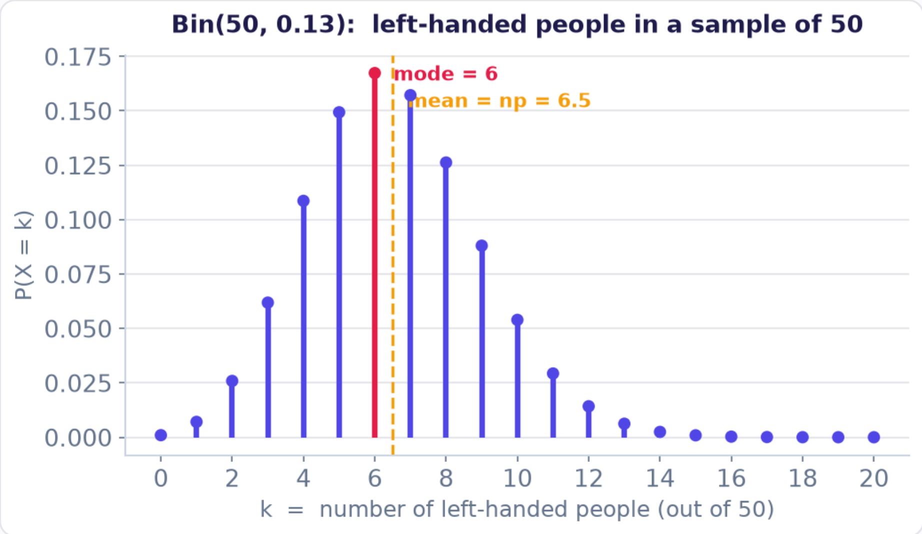
You will almost never expand the formula by hand for large n. Learn these two tools cold — the crucial choice is **exactly equal (pdf / FALSE)** vs **cumulative "≤" (cdf / TRUE)**.

You want	Excel	Graphing calculator (2nd → DISTR)
$P(X = k)$ (exactly)	`=BINOM.DIST(k; n; p; FALSE)`	`binompdf(n, p, k)`
$P(X \leq k)$ (cumulative)	`=BINOM.DIST(k; n; p; TRUE)`	`binomcdf(n, p, k)`

Everything else is built from these two by translating the wording into ≤ statements (see the table in Part 6).

WORKED — LEFT-HANDEDNESS: X ~ BIN(50, 0.13)

13% of people are left-handed; we pick 50 at random. X = number who are left-handed.



The distribution: mean 6.5, mode at $k = 6$.

Sub-question	How to set it up	Answer
Expected number	$np = 50 \times 0.13$	**6.5**
Standard deviation	$\sqrt{(np(1-p))} = \sqrt{(50 \cdot 0.13 \cdot 0.87)}$	≈ **2.38**
Mode	tallest needle on the graph (near np)	**6**
Exactly 10	$P(X=10)$ = `binompdf(50,0.13,10)`	≈ 0.043
At least 10	$1 - P(X \le 9) = 1 -$ `binomcdf(50,0.13,9)`	≈ 0.115
At least 5 but at most 9	$P(X\le9) - P(X\le4)$ = `binomcdf(…,9) − binomcdf(…,4)`	≈ 0.667

The "how many to check" twist: to be 99% sure of at least 100 left-handers you'd increase n until $P(X \ge 100) \ge 0.99$, i.e. solve $1 - P(X \le 99) \ge 0.99$ for n by trying values on the calculator. Same idea as the survey problem below.

WORKED — THE SURVEY PROBLEM (READING THREE QUESTION TYPES)

You send letters to entrepreneurs; each responds independently with probability = response rate. R = number returned.

- **Given n=400, p=0.6 — probability at least 250 return?** R~Bin(400, 0.6). Want $P(R \geq 250) = 1 - P(R \leq 249)$ = 1 − `binomcdf(400, 0.6, 249)`. (Here n and p are known → plug in.)
- **What p makes $P(R \geq 250)$ = 0.95, with n=400?** Now p is the unknown — increase p until the cumulative condition is met (trial and error, or solve).
- **How many letters n to get 250 back with 95% probability, p=0.6?** Now n is the unknown — increase n until $P(R \geq 250) \geq 0.95$.

The lesson: the same binomial model appears three times; which quantity is unknown (n, p, or a probability) tells you whether to plug in directly or to search for the value that satisfies the condition.

HOW TO THINK — ANY BINOMIAL QUESTION

1. **Confirm it's binomial (BINS)** and state $X \sim \text{Bin}(n, p)$ with the actual numbers.
2. **Identify the target.** Is a summary asked (mean np, sd $\sqrt{(np(1-p))}$, mode) or a probability?
3. **Translate the words into ≤.** "Exactly" → pdf; "at most / at least / between" → build from cdf (Part 6 table).
4. **Use the complement for "at least".** $P(X \geq k) = 1 - P(X \leq k-1)$.
5. **If the unknown is n or p** (not a probability), set up the ≥ 0.95 / ≥ 0.99 condition and search for the value that meets it.

6 Exam strategy & formula card

YOUR ONE-PAGE REFERENCE FOR THE WHOLE CHAPTER

Translating words into inequalities — the table that saves marks

Most errors are not calculation errors; they are translation errors. Keep this handy:

Words in the question	Event	How to compute (discrete)
"exactly k"	$X = k$	pdf value / `binompdf`
"at most k", "no more than", "k or fewer"	$X \le k$	$F(k)$ / `binomcdf(…,k)`
"fewer than k", "less than"	$X < k$	$F(k-1)$ / `binomcdf(…,k-1)`
"at least k", "k or more", "no fewer than"	$X \ge k$	$1 - F(k-1)$
"more than k", "exceeds"	$X > k$	$1 - F(k)$
"between a and b" (inclusive)	$a \le X \le b$	$F(b) - F(a-1)$

THE TWO MISTAKES THAT COST THE MOST MARKS

- **Off-by-one on "at least".** $P(X \ge k) = 1 - P(X \le k-1)$, not $1 - P(X \le k)$. Draw the needles and count if unsure.
- **Variance formula order.** It's $E(X^2) - [E(X)]^2$, the mean of the squares minus the square of the mean — never the other way round.

Master decision flow

WHEN YOU MEET A DISCRETE-RANDOM-VARIABLE QUESTION

1. **Set the scene:** name the experiment, define X + units, confirm discrete.
2. **Is it a named model?** Two outcomes repeated a fixed independent n times → **binomial**, use its formulas. Otherwise work from the given probability table.
3. **What's asked?**
   - A single probability → translate words → inequalities (table above) → add probabilities or use pdf/cdf.
   - A cumulative / median / quantile → walk the $F(x)$ column.
   - A centre → $E(X) = \Sigma\, m_i P$ (or np).
   - A spread → $\mathrm{Var} = E(X^2) - [E(X)]^2$ (or $np(1-p)$), then √ for σ.
4. **Sanity-check** the answer: probabilities in [0,1], mean inside the outcome range, complement used for long "at least" tails.

Formula card

Concept	Formula	Notes
Distribution rules	$0 \le P \le 1$ and $\Sigma P = 1$	use "sums to 1" to fill gaps
CDF	$F(x) = P(X \le x)$	step function; right-continuous
Interval	$P(a < X \le b) = F(b) - F(a)$	mind < vs ≤ at the ends
Median	smallest x with $F(x) \ge 0.5$	"first to reach", not "closest"
Mean	$E(X) = \Sigma\ m_i\ P(X{=}m_i)$	weighted average
Variance	$Var(X) = E(X^2) - [E(X)]^2$	$= \Sigma(m_i - \mu)^2 P$
St. deviation	$\sigma = \sqrt{Var(X)}$	same units as X
Bernoulli	$E = p, \quad Var = p(1-p)$	single 0/1 trial
Binomial pmf	$P(X{=}k) = C(n,k)\ p^k(1-p)^{n-k}$	$X \sim Bin(n,p)$
Binomial mean/var	$E = np, \quad Var = np(1-p)$	sum of n Bernoullis

FINAL WORD

Nearly every question in this chapter is one of a handful of moves: complete a table, add up the right outcomes, climb the cumulative column, or plug into a mean/variance formula. The hard part is almost never the arithmetic — it's **translating the sentence into the right event or formula**. Define X clearly, turn the words into inequalities, and reach for the complement whenever you see "at least". Do that and the rest is bookkeeping.